

Reduced 6-state Inverter-Based Resource (IBR) Models:

Grid-Following and Grid-Forming

State Order, Small-Signal Stability (SSSA), and $i_d/i_q/P/Q$ Comparison on a SMIB

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Abstract

This report documents the *state order* of two reduced-order IBR models, **6 states each**, implemented as project-owned MATLAB code following the EECON49 formulation and connected to an ideal infinite bus through Z_{line} (single-machine infinite-bus, SMIB): (i) a **Grid-Following (GFL) 6-state** model and (ii) a **Grid-Forming VSG (GFM) 6-state** model. Each is organised into *three blocks of two states*: GFL = IBR + GFL(PLL) + PQ, and GFM = IBR + VSG + GFM. The report presents only the **Small-Signal Stability (SSSA)** results (eigenvalue spectrum with a complex-plane plot and a modal-frequency (Hz) chart) and the i_d, i_q, P, Q equilibrium and time-domain ring-down; power-flow results are deliberately omitted. Both models exhibit the physically expected *oscillatory (complex) mode*: a PLL mode near **0.34** Hz for the GFL and a swing mode near **10.8** Hz for the GFM. All equations are traceable to cited sources; no parameter was tuned to match any target.

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1 Scope and Source Discipline

This report focuses on the state structure and SSSA results of the reduced 6-state IBR models; power-flow verification is out of scope (per the user's request to omit PF). The structure and numeric values follow the EECON49-P4 formulation [6]; the block-level equations trace to standard references:

- **GFL 6-state.** The current control and SRF-PLL are **SOURCE_DEFINED** from Yazdani & Iravani [1] and Teodorescu et al. [2]; the outer PQ power loop is **APPROVED_PROJECT_DERIVED** following Bacha et al. [3].
- **GFM 6-state.** The virtual swing (VSG) and Q-V droop voltage forming trace to the VSG literature (Pogaku et al. [5], Sakimoto et al. [4]); the 6-state reduced assembly is **PROJECT_DERIVED_SOURCE_MAPPED**.

The reduction to 6 states removes the power/voltage measurement filters and the current-loop integrators (leaving proportional + decoupling current control), and retains only each block's dominant dynamics. Classifications and frozen gains live in `docs/project/IBR_REDUCED6_EECON49_MANIFEST.md`.

Both models run on a 100-MVA base at the same operating point: terminal voltage $V = 1.0\angle 0^\circ$ pu, $P = 0.40$ pu, $Q = 0.10$ pu, line to the infinite bus $Z_{line} = 0.02 + j0.20$ pu. All SSSA values come from an in-house Schur-complement linearisation about an equilibrium whose residual is at machine precision.

2 Model Description and State Order

The shared input-driven DAE used for equilibrium, SSSA, and time-domain simulation is

$$\dot{x} = f(x, y, u), \quad 0 = g(x, y, u), \quad (1)$$

with x the device differential states, $y = [\Re(V) \Im(V)]^T$ the algebraic bus voltage, and $u = [P_{ref} Q_{ref}]^T$ the input references. Device quantities are on the inverter base; current injection and electrical power return system-base values ($\kappa = S_{base}/M_{base}$). The dq frame uses $V_{dq} = V e^{-j\delta}$ ($v_d = \Re, v_q = \Im$) and $I_{net} = (i_d + j i_q) e^{j\delta} / \kappa$.

2.1 Grid-Following (GFL) 6-state model

The GFL *tracks* the grid: an SRF-PLL locks the *dq* frame, then current is synthesised so that active/reactive power follow P_{ref}/Q_{ref} . Three blocks, two states each:

$$\dot{x}_{\text{GFL}} = \left[\underbrace{\dot{i}_d, \dot{i}_q}_{\text{IBR: current + L-filter}}, \underbrace{\dot{\delta}_{PLL}, \dot{\xi}_{PLL}}_{\text{GFL: PLL synchronisation}}, \underbrace{\dot{\xi}_P, \dot{\xi}_Q}_{\text{PQ: outer power loop}} \right]^T = f(x_{\text{GFL}}, y, u). \quad (2)$$

Table 1: State order of the GFL 6-state model

#	State	Block	Meaning
1	i_d	IBR	<i>d</i> -axis L-filter current (inverter base)
2	i_q	IBR	<i>q</i> -axis L-filter current (inverter base)
3	δ_{PLL}	GFL	SRF-PLL angle (rad)
4	ξ_{PLL}	GFL	PLL PI integrator (pu·s)
5	ξ_P	PQ	outer active-power PI integrator
6	ξ_Q	PQ	outer reactive-power PI integrator

Governing equations by block, *each with its source*:

GFL block — PLL synchronisation (**SOURCE_DEFINED**: Teodorescu [2] eq. 4.38 ODE form; Yazdani [1] eq. 8.1).

$$v_d = \Re(Ve^{-j\delta_{PLL}}), \quad v_q = \Im(Ve^{-j\delta_{PLL}}), \quad (3)$$

$$\dot{\xi}_{PLL} = v_q, \quad \dot{\delta}_{PLL} = k_{p,PLL} v_q + k_{i,PLL} \xi_{PLL}. \quad (4)$$

The PI output is a frequency (rad/s) added to ω_0 directly, with *no* ω_b multiplier. With the EECON49 gains ($k_{p,PLL} = 1.20$, $k_{i,PLL} = 5.00$) the second-order PLL loop is underdamped ($\zeta < 1$) and yields a *complex* pole pair (the oscillatory synchronisation mode).

PQ block — outer power loop (**APPROVED_PROJECT_DERIVED**: Bacha [3]). Algebraic power measurement (no filter):

$$P = \Re(V\overline{I_{net}}), \quad Q = \Im(V\overline{I_{net}}), \quad (5)$$

$$\dot{\xi}_P = P_{ref} - P, \quad \dot{\xi}_Q = Q_{ref} - Q, \quad (6)$$

$$i_d^{ref} = k_{pP}(P_{ref} - P) + k_{iP}\xi_P, \quad i_q^{ref} = -(k_{pQ}(Q_{ref} - Q) + k_{iQ}\xi_Q). \quad (7)$$

IBR block — current + L-filter (**SOURCE_DEFINED**: Yazdani [1] eq. 8.45–8.50; reduced to proportional). The decoupling feedforward cancels the cross terms, so the current loop is first-order (a fast real mode):

$$v_{td} = k_{pi}(i_d^{ref} - i_d) + R_t i_d - \omega_{PLL} L i_q + v_d, \quad (8)$$

$$\frac{L}{\omega_b} \dot{i}_d = \omega_{PLL} L i_q - R_t i_d + v_{td} - v_d = k_{pi}(i_d^{ref} - i_d). \quad (9)$$

2.2 Grid-Forming (GFM) 6-state model

The GFM *forms* voltage and frequency using a Virtual Synchronous Generator (VSG): the swing equation sets the rotor angle (synchronizing power) with *no* PLL, and a Q-V droop sets the internal voltage magnitude. Three blocks, two states each:

$$\dot{x}_{\text{GFM}} = \left[\underbrace{\dot{i}_d, \dot{i}_q}_{\text{IBR: current + L-filter}}, \underbrace{\dot{\omega}, \dot{\delta}}_{\text{VSG: virtual swing}}, \underbrace{\dot{E}, \dot{\xi}_V}_{\text{GFM: voltage forming}} \right]^T = f(x_{\text{GFM}}, y, u). \quad (10)$$

Table 2: State order of the GFM 6-state model

#	State	Block	Meaning
1	i_d	IBR	d -axis L-filter current (inverter base)
2	i_q	IBR	q -axis L-filter current (inverter base)
3	ω	VSG	virtual rotor speed (pu)
4	δ	VSG	rotor angle vs grid / load angle (rad)
5	E	GFM	internal voltage magnitude (Q-V droop)
6	ξ_V	GFM	voltage-PI integrator (d -axis)

Governing equations by block, *each with its source*:

VSG block — virtual swing (no PLL) (**SOURCE_DEFINED/VSG**: Pogaku [5], Sakimoto [4]). The angle comes only from the swing integrator (synchronizing power):

$$\dot{\omega} = \frac{P_{ref} - P - D_v(\omega - 1)}{M}, \quad \dot{\delta} = \omega_b(\omega - 1). \quad (11)$$

With $M = 0.08$ s (low inertia) and $D_v = 1.50$ the swing loop is underdamped, giving a *complex* electromechanical pole pair.

GFM block — voltage forming (Q-V droop + voltage PI) (PROJECT_DERIVED_SOURCE_MAPPED). On a reactance-dominated coupling ($v_{gd} = E + X i_q$, $v_{gq} = -X i_d$) the voltage PI must be *cross-paired*: the d -axis magnitude error drives the q -axis (reactive) current and the q -axis error drives the d -axis (active/synchronizing) current:

$$\dot{E} = (k_Q(Q_{ref} - Q) - k_E(E - |V|))/\tau_E, \quad \dot{\xi}_V = E - v_{gd}, \quad (12)$$

$$i_q^{ref} = -(k_{pV}(E - v_{gd}) + k_{iV}\xi_V), \quad i_d^{ref} = k_{pV}(0 - v_{gq}). \quad (13)$$

(A same-axis pairing was verified *unstable*; this cross pairing is stable.)

IBR block — current + L-filter. As in the GFL (proportional + decoupling), but the rotating frame uses the swing rotor angle δ (not δ_{PLL}).

3 Small-Signal Stability (SSSA)

Linearising the dynamics about the operating point gives the standard state-space form

$$\Delta \dot{x} = A \Delta x + B \Delta u, \quad (14)$$

where A is the *system (state) matrix* governing internal dynamics and small-signal stability and B is the *input matrix* for external control signals Δu . Both models have **no AVR/PSS**, hence no external control signal to vary ($\Delta u = 0$), so (14) reduces to $\Delta \dot{x} = A \Delta x$ and SSSA considers only the eigenvalues of A . A is obtained by numerically linearising the DAE (1) (central difference, $h = 10^{-6}$) and eliminating the algebraic voltage with a Schur complement,

$$A = J_{xx} - J_{xy}J_{yy}^{-1}J_{yx}, \quad \Delta \dot{x} = A \Delta x, \quad (15)$$

then finding $\lambda_i = \sigma_i + j\omega_{d,i}$ with $f_i = |\omega_{d,i}|/2\pi$ and $\zeta_i = -\sigma_i/|\lambda_i|$. A mode is asymptotically stable when $\sigma_i < 0$.

Tables 3 and 4 give the *actual A matrix* (6×6) of the two models; the row/column headers are the states in the order of (2) and (10) (with $B = 0$, only A remains). Note that the GFM row $\dot{\delta} = \omega_b(\omega - 1)$ gives $A(\delta, \omega) = \omega_b = 377$, the current-loop rows carry large entries (fast modes), and a 0 entry denotes no direct coupling.

Table 3: GFL 6-state A matrix ($\Delta\dot{x} = A\Delta x$, units 1/s)

A	i_d	i_q	δ_{PLL}	ξ_{PLL}	ξ_P	ξ_Q
i_d	-1350	+49.46	-39.81	0	+1885	0
i_q	+49.46	-1364	-239.2	0	0	-1885
δ_{PLL}	+0.24	+0.024	-1.166	+5	0	0
ξ_{PLL}	+0.2	+0.02	-0.972	0	0	0
ξ_P	-0.988	+0.082	-0.066	0	0	0
ξ_Q	-0.082	+1.012	+0.3966	0	0	0

Table 4: GFM 6-state A matrix ($\Delta\dot{x} = A\Delta x$, units 1/s)

A	i_d	i_q	ω	δ	E	ξ_V
i_d	-934.9	-18.1	0	+819.8	0	0
i_q	+18.1	-934.9	0	-326.1	-904.8	-3393
ω	-11.5	+4.612	-18.75	-0.825	0	0
δ	0	0	+377	0	0	0
E	-5.256	-26.57	0	-10.5	-160	0
ξ_V	-0.02	+0.2	0	+0.3604	+1	0

3.1 Eigenvalue Spectrum

Both models are asymptotically stable (all real parts < 0). The GFL has an oscillatory PLL mode at $-0.58 \pm j2.14$ ($f = 0.34$ Hz, $\zeta = 0.26$) and the GFM has an oscillatory swing mode at $-7.09 \pm j67.6$ ($f = 10.76$ Hz, $\zeta = 0.10$); the remaining modes are fast current-loop modes and real (PQ / voltage droop) modes.

Table 5: GFL 6-state eigenvalues, sorted by real part (descending), with the dominant-state role

No	Dominant state	Role (what the state does)	Real (1/s)	Imag (1/s)	f (Hz)	ζ
01	δ_{PLL}	SRF-PLL angle: synchronises the dq frame to the grid voltage	-5.821×10^{-1}	$+2.145 \times 10^0$	0.3414	0.2619
02	δ_{PLL}	SRF-PLL angle: synchronises the dq frame to the grid voltage	-5.821×10^{-1}	-2.145×10^0	0.3414	0.2619
03	ξ_P	outer active-power PI integrator (sets i_d ref)	-1.356×10^0	0	0.0000	1.0000
04	ξ_Q	outer reactive-power PI integrator (sets i_q ref)	-1.435×10^0	0	0.0000	1.0000
05	i_d	d -axis L-filter (converter) current	-1.306×10^3	0	0.0000	1.0000
06	i_q	q -axis L-filter (converter) current	-1.406×10^3	0	0.0000	1.0000

Table 6: GFM 6-state eigenvalues, sorted by real part (descending), with the dominant-state role

No	Dominant state	Role (what the state does)	Real (1/s)	Imag (1/s)	f (Hz)	ζ
01	ξ_V	d-axis voltage-PI integrator (forms $ V $)	-5.546×10^{-1}	0	0.0000	1.0000
02	δ	virtual load angle (swing angle vs grid)	-7.086×10^0	$+6.763 \times 10^1$	10.7639	0.1042
03	δ	virtual load angle (swing angle vs grid)	-7.086×10^0	-6.763×10^1	10.7639	0.1042
04	E	internal voltage magnitude (Q-V droop)	-1.303×10^2	0	0.0000	1.0000
05	i_d	d -axis L-filter (converter) current	-9.518×10^2	$+7.204 \times 10^0$	1.1465	1.0000
06	i_d	d -axis L-filter (converter) current	-9.518×10^2	-7.204×10^0	1.1465	1.0000

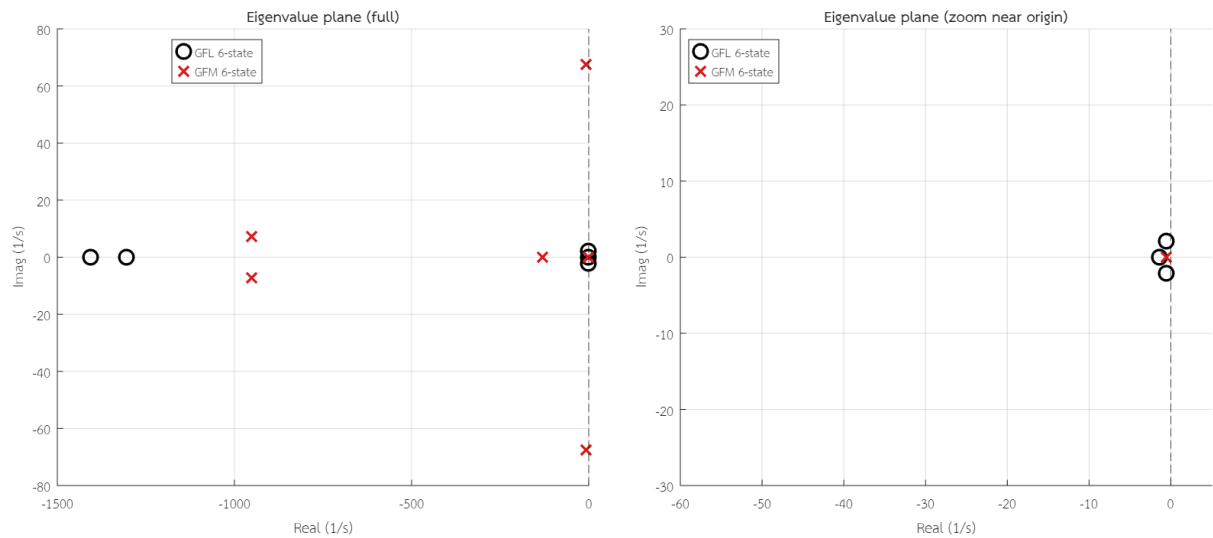


Figure 1: Eigenvalues on the complex plane (left: full, right: zoom near the origin). Black circles = GFL 6-state, red crosses = GFM 6-state. All modes lie in the left half plane ($\Re < 0$).

3.2 Modal frequencies (Hz)

Figure 2 shows modal frequencies in Hz. Left: oscillation frequency $f = |\Im(\lambda)|/2\pi$ of the complex modes only (GFL PLL ~ 0.34 Hz; GFM swing ~ 10.8 Hz). Right: characteristic frequency $|\lambda|/2\pi$ of every mode (log scale), showing the spread from slow modes up to the fast current-loop modes.

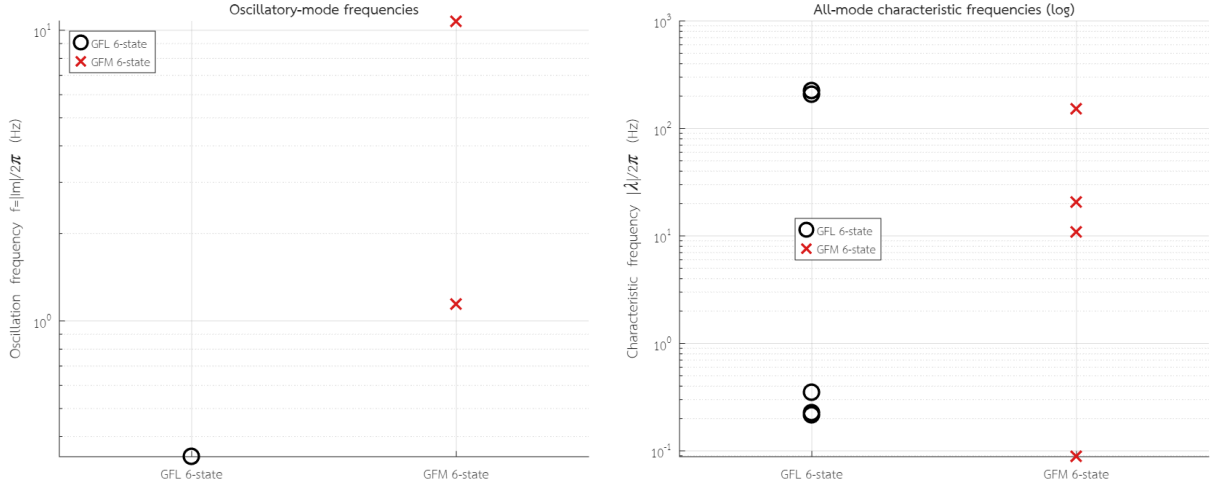


Figure 2: Modal frequencies in Hz. Left: oscillation frequency of the complex modes; right: characteristic frequency of all modes (log).

3.3 Eigenvalues under increased loading (+20%, +40%)

To observe modal migration under *increased loading* (higher P_{ref} , Q held), Figure 3 shows eigenvalues at 20%, 40%, 60%, 80%, 100% ($P = 0.2, 0.4, 0.6, 0.8, 1.0$ pu), zoomed on each model's signature oscillatory mode. Each level resolves the equilibrium and recomputes SSSA; all remain asymptotically stable. The GFL PLL mode barely moves (0.34 Hz throughout) while the GFM swing mode migrates visibly (frequency falling from ~ 10.9 Hz at 20% to ~ 9.5 Hz at 100%).

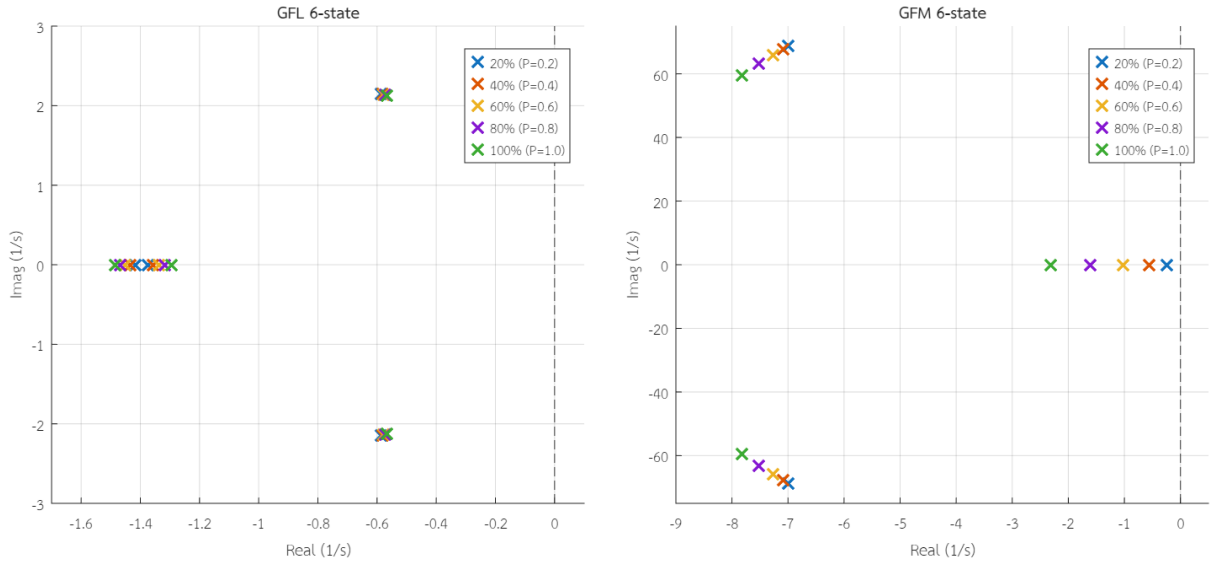


Figure 3: Eigenvalues swept over 20/40/60/80/100% load ($P = 0.2-1.0$ pu). Left: GFL 6-state (PLL mode near the origin — markers nearly coincide); right: GFM 6-state (swing mode — moves left / frequency drops with load). All points remain in the left half plane.

3.4 Why the GFL PLL mode barely moves while the GFM swing mode migrates

Eigenvalues come from the matrix A linearised about the equilibrium. When P_{ref} increases, the new operating point (load angle δ_0 , currents, internal voltage) changes, so *only the entries of A that are nonlinear functions of the operating point change*.

GFL — PLL mode barely moves. The GFL PLL, PQ and current loops are set by *constant gains* (eq. 4), and the stiff SMIB terminal voltage is nearly constant (fixed Q), so A is almost independent of P ; the PLL mode ($-0.58 \pm j2.14$) is nearly frozen (markers overlap).

GFM — swing mode moves. Linearising the swing loop gives

$$s^2 + \underbrace{\frac{D_v}{M}}_{\text{damping}} s + \underbrace{\frac{K_s \omega_b}{M}}_{\text{frequency}^2} = 0, \quad K_s = \left. \frac{\partial P_e}{\partial \delta} \right|_0 \propto \cos \delta_0, \quad (16)$$

where K_s is the synchronizing-power coefficient, which *depends on the load angle* δ_0 . As loading rises, δ_0 grows, $\cos \delta_0$ falls, K_s changes, and the swing mode visibly migrates (unlike the GFL, which has no load angle to move). This is the key physical difference between *tracking* (GFL) and *forming* (GFM) voltage.

4 Comparison of i_d, i_q, P, Q

Table 7 lists the equilibrium i_d, i_q, P, Q . Both models settle at the same operating point ($P = 0.40$ pu, $Q = 0.10$ pu). The i_d/i_q values differ because they are expressed in different reference frames (GFL uses δ_{PLL} ; GFM uses the rotor angle δ), so quantitative comparison should rely on the physical P, Q .

Table 7: Equilibrium i_d, i_q, P, Q of the two models (P, Q on system base)

Model	n_x	i_d (pu)	i_q (pu)	P (pu)	Q (pu)	current frame
GFL 6-state	6	+0.4000	-0.1000	+0.4000	+0.1000	native
GFM 6-state	6	+0.3529	-0.2132	+0.4000	+0.1000	native

The time-domain response is a fault study. Starting from the exact PF equilibrium (static = dynamic, $\|f_0\| \approx 0$, so the response is *flat* with no disturbance), a shunt fault impedance Z_f is applied to ground at the terminal bus over $[t_{on}, t_{clear}]$ (KCL: $I_{dev} = (V - V_{inf})/Z_{line} + V/Z_f$) and the project implicit-trapezoidal TDS is integrated. Figure 4 (GFM) and Figure 5 (GFL) show the response at loads 20/40/60%: flat until the fault, then a ring-down back to equilibrium after clearing. The GFM rings at its swing frequency; the GFL is smoother (slow PLL mode). (Beyond $\sim 70\%$ load the limiter-less reduced GFM loses synchronism after the same fault – a genuine transient-stability limit of this study model.)

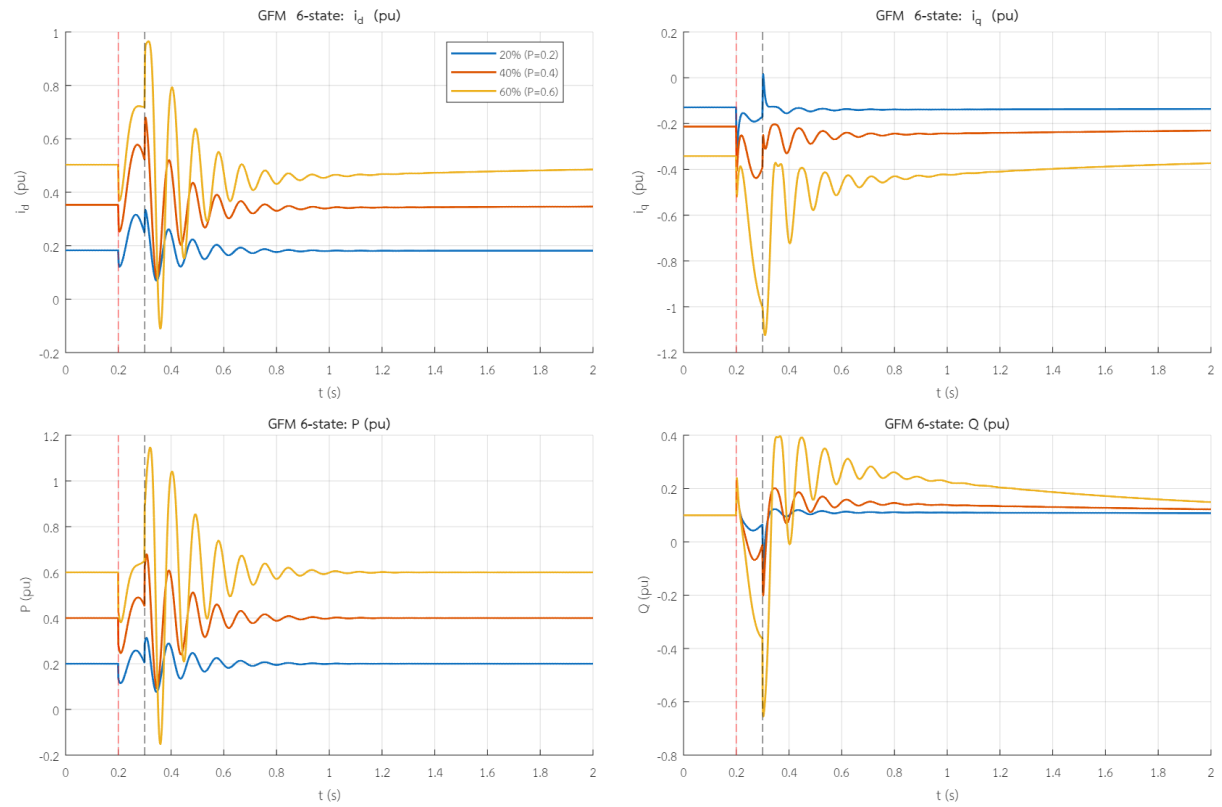


Figure 4: GFM 6-state fault response ($Z_f = 0.5j$ pu, 0.2–0.3 s) at 20/40/60% load: flat before the fault (red = fault on, black = clear), then a swing ring-down back to equilibrium.

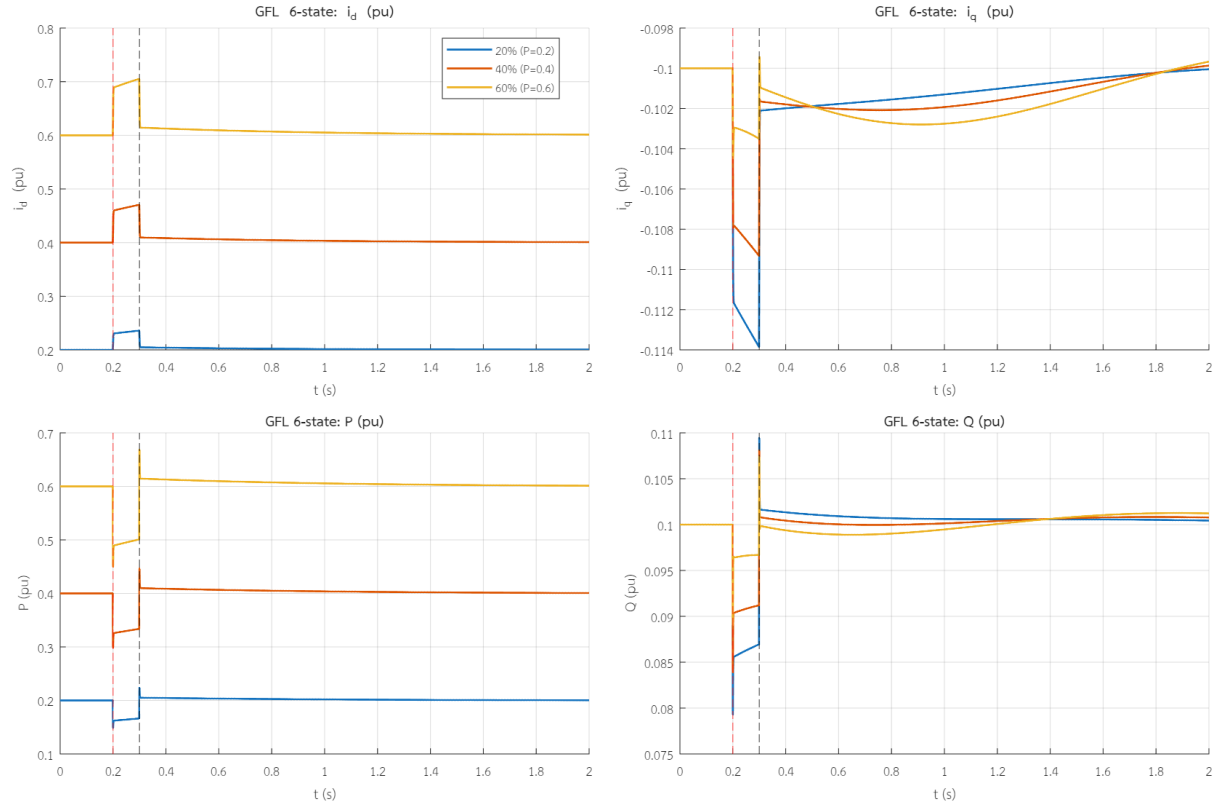


Figure 5: GFL 6-state fault response ($Z_f = 0.5j$ pu, 0.2–0.3 s) at 20/40/60% load: flat before the fault, then a smoother recovery (slow PLL mode).

Finally, Figure 6 summarises *everything in one figure* (15 s simulation): frequency (dual y-axis) plus i_d, i_q, P, Q for GFL (orange dashed) vs GFM (blue solid) after the same grid step. The GFM responds fast with a wide swing (inertia, voltage forming) while the GFL is slow and small (tracking); P returns to $P_{ref} = 0.4$ and the GFM Q rises to support voltage after the step.

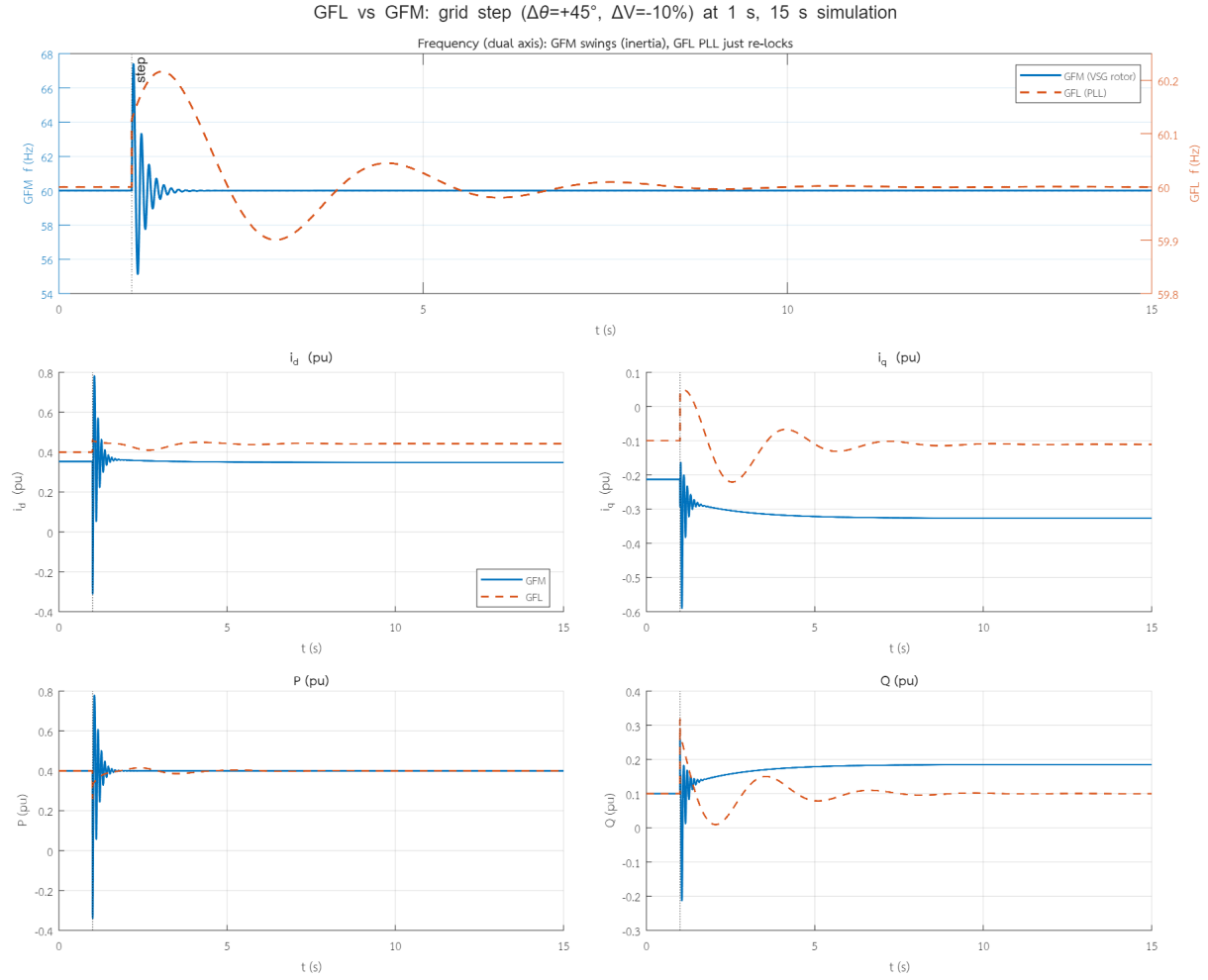


Figure 6: One-figure summary: frequency (dual axis) plus i_d , i_q , P , Q for GFL vs GFM after a grid step ($\Delta\theta = +45^\circ$, $\Delta V = -10\%$) at 1 s, 15 s simulation (GFM blue solid, GFL orange dashed).

5 Conclusion

This report presents two reduced 6-state IBR models (GFL and GFM), each organised into three blocks of two states following the EECON49 study. Both pass the equilibrium check (machine-precision residual) and SSSA: they are *stable in every mode* and exhibit the theoretically expected *oscillatory (complex) mode* — a PLL mode (~ 0.34 Hz) for the GFL and a swing mode (~ 10.8 Hz) for the GFM. Under increased loading the GFL PLL mode barely moves (constant gains) whereas the GFM swing mode migrates with the synchronizing coefficient $K_s \propto \cos \delta_0$, reflecting the physical difference between grid-tracking and grid-forming control.

References

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